

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks Using Large-Scale Charging Session Data.

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DATA LANDSCAPE & PREPROCESSING

Data Sources

UrbanEV Dataset

Large-scale EV charging station operational dataset collected from charging infrastructure in China.

Contains station-level occupancy, charging volume, session duration, and pricing information.

Recorded at 5-minute intervals across multiple charging stations.

Used as the primary dataset for demand forecasting, tariff optimization, and monitoring.

ACN Dataset

Public EV charging session dataset collected from charging networks in the United States.

Contains connection, charging completion, and disconnection timestamps along with energy delivered.

Used to analyze user charging behavior, charger utilization efficiency, and idle occupancy patterns.

Data Preprocessing

UrbanEV

Converted wide-format station matrices into long-format observations.

Merged occupancy, volume, duration, and pricing datasets using Timestamp and Station ID.

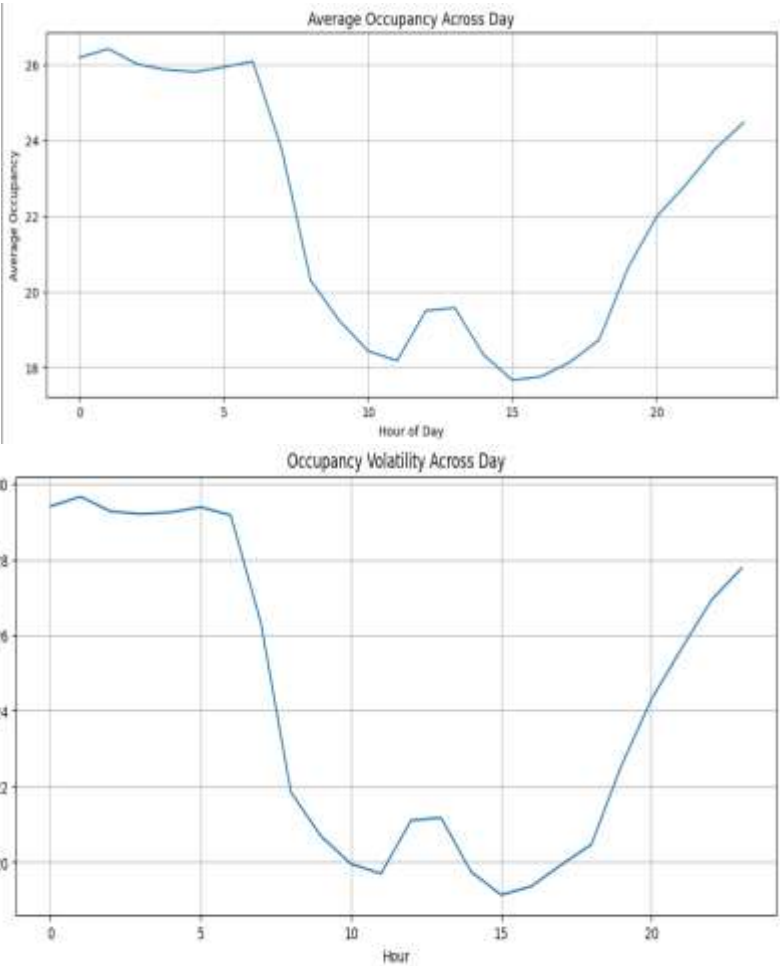
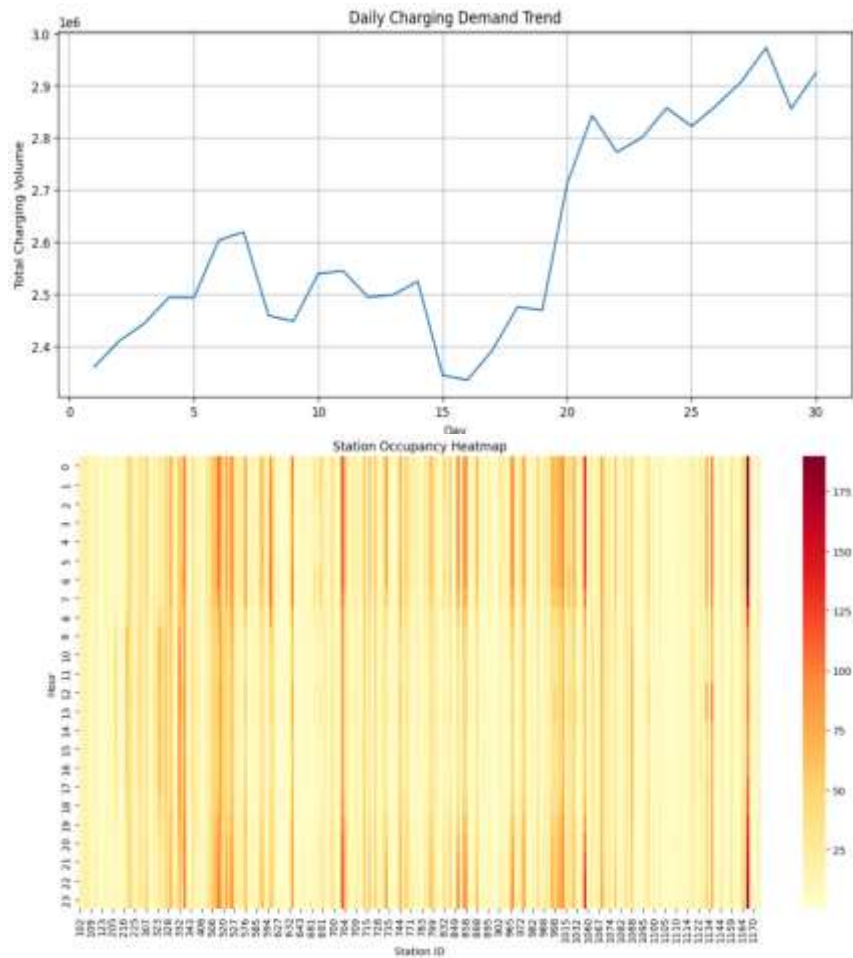
Created a unified analytical dataset for modelling and optimization.

ACN

Removed incomplete charging sessions containing missing operational timestamps.

Retained only critical session-level variables required for behavioral analysis.

EDA & DEMAND BEHAVIOR INSIGHTS



Contrary to conventional assumptions, the network's peak utilization occurred during overnight hours. Therefore, peak periods were identified directly from data rather than predefined using human assumptions.

- Key Findings**
- Long-Term Demand Trend**
 - Charging demand increased over the month
 - Demand dip observed around Day 15–16
 - Indicates evolving network utilization
- Intraday Charging Behavior**
 - Highest occupancy observed during late-night and early-morning hours
 - Lowest occupancy during midday
- Congestion Characteristics**
 - Significant variation across charging stations
 - Some stations consistently experience high utilization
- Volatility Analysis**
 - Occupancy variability highest during high-demand periods
 - Stable utilization observed during midday

DEMAND PREDICTION AGENT

Predict future charging demand and station occupancy using historical operational features.

Features Used : Volume, Duration, Price, Charger Utilization Rate , Hour of Day

Target : Occupancy

Models Evaluated

Linear Regression - Baseline predictive model

Random Forest - Non-linear ensemble model

Model	MAE	RMSE	R ²
Linear Regression	4.498	7.557	0.907
Random Forest	0.796	1.857	0.994

Interpretation

Linear Regression captured general trends.

Random Forest captured complex non-linear operational relationships.

Final Demand Prediction Agent used Random Forest.

Important Decision

Occupancy Density and Queue Length Proxy were excluded from modelling because they were derived from occupancy and would introduce target leakage.

DYNAMIC TARIFF PRICING AGENT

Translate demand forecasts into adaptive pricing decisions.

Pricing Logic

Surge Pricing : Applied when, Predicted Utilization > 80% ,Price Increase: +35%

Discount Pricing : Applied when, Predicted Utilization < 30%, Price Reduction: -5%

Customer Response Simulation

Assumptions:

High prices slightly reduce demand

Discounts encourage additional off-peak charging

Used as a proxy for demand elasticity

Metric	Result
Revenue Gain	+2.44%
Off-Peak Uplift	+8.0%
Utilization Before	0.295
Utilization After	0.323

Interpretation

Dynamic pricing successfully redistributed charging demand toward low-utilization periods while improving both revenue and charger utilization.

MONITORING & LEARNING AGENT

Evaluate the effectiveness of pricing decisions and create a feedback loop.

Metrics Tracked

Wait Time Reduction

Proxy for congestion reduction

Customer Response Rate

Measures demand shift after pricing changes

Pricing Efficiency Score

Revenue generated per unit charging activity

Metric	Result
Wait Time Reduction	12.0%
Customer Response Rate	5.24%
Pricing Efficiency Gain	-2.66%

Interpretation

Congestion reduced significantly

Users responded positively to pricing incentives

Slight efficiency tradeoff observed due to off-peak discounts

Key Insight

Dynamic pricing creates a tradeoff between operational efficiency and revenue efficiency. The proposed system prioritizes congestion mitigation while maintaining positive revenue growth.

BUSINESS, OPERATIONAL & POLICY IMPLICATIONS

Business Impact

- Increased charging revenue
- Improved charger utilization
- Better asset efficiency

Operational Impact

- Reduced congestion
- Lower estimated waiting times
- Improved load balancing across charging periods

Policy Implications

- Supports sustainable EV infrastructure management
- Encourages off-peak charging behavior
- Reduces need for immediate infrastructure expansion

Demand Prediction Agent



Tariff Pricing Agent



Customer Response



Monitoring & Learning Agent



Continuous Optimization

CONCLUSION

Contributions

- Unified multi-source EV charging datasets
- Built Demand Prediction Agent
- Built Dynamic Pricing Agent
- Built Monitoring & Learning Agent
- Developed closed-loop Agentic AI framework

Final Results

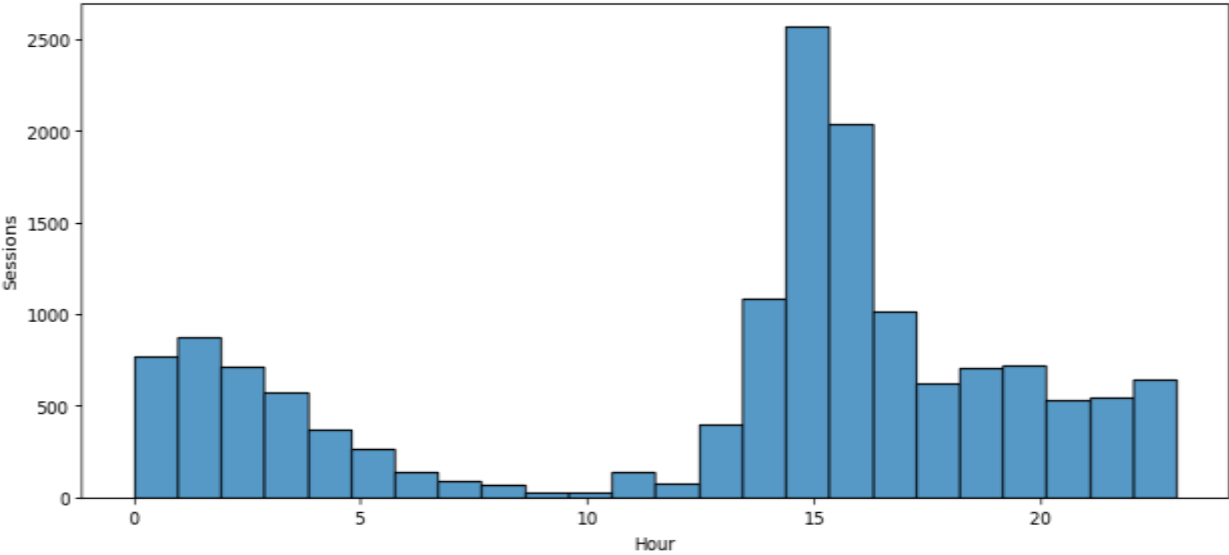
- $R^2 = 0.994$
- Revenue Gain = 2.44%
- Off-Peak Uplift = 8%
- Wait Time Reduction = 12%

Future Work

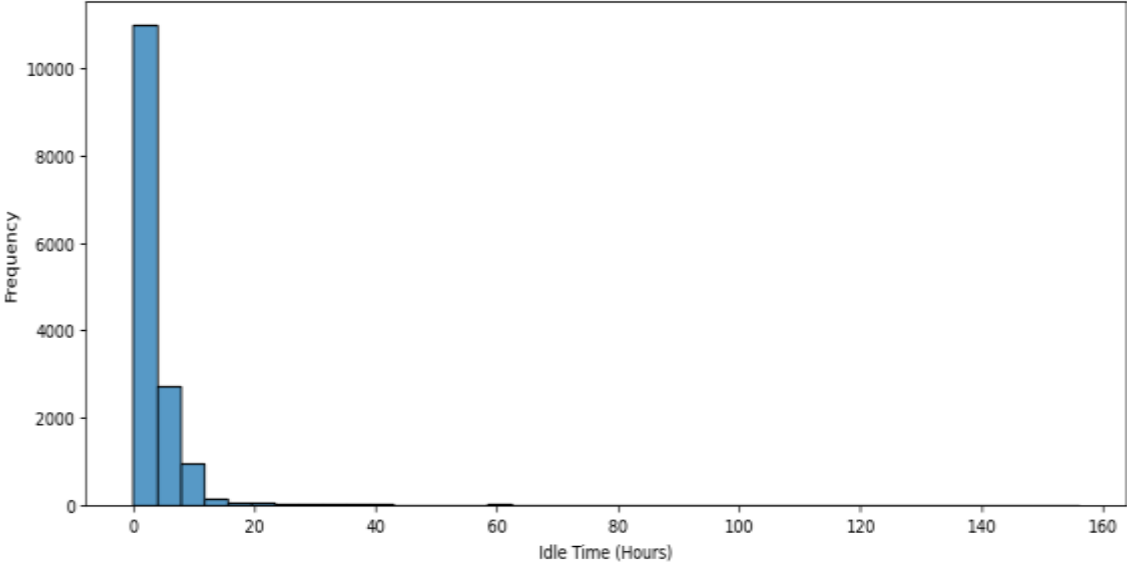
- Real-time deployment
- Reinforcement learning pricing policies
- Weather and event-aware forecasting
- Multi-city charging network optimization

APPENDIX

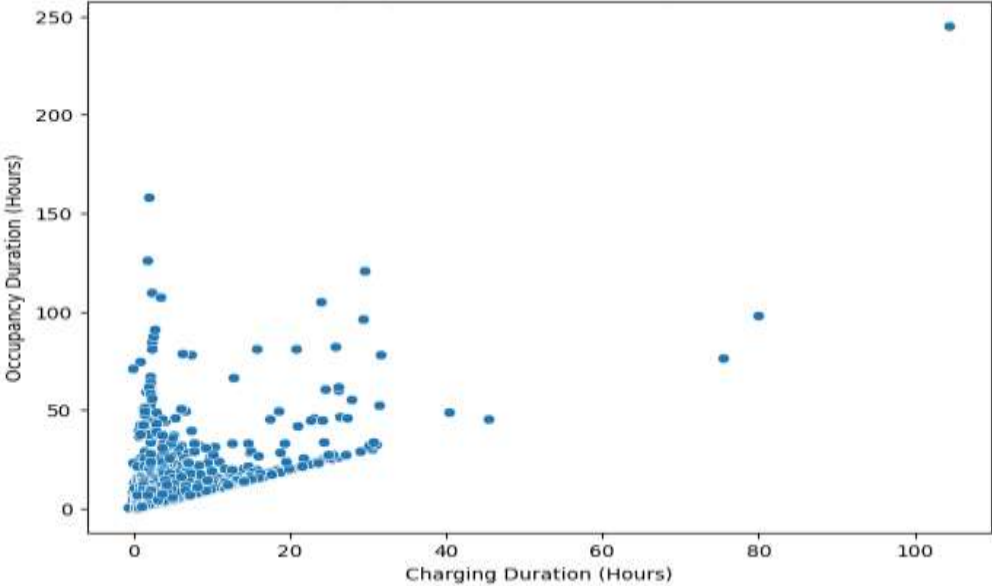
Peak Charging Hours



Idle Time After Full Charge



Charging Duration vs Occupancy Duration



Thank You